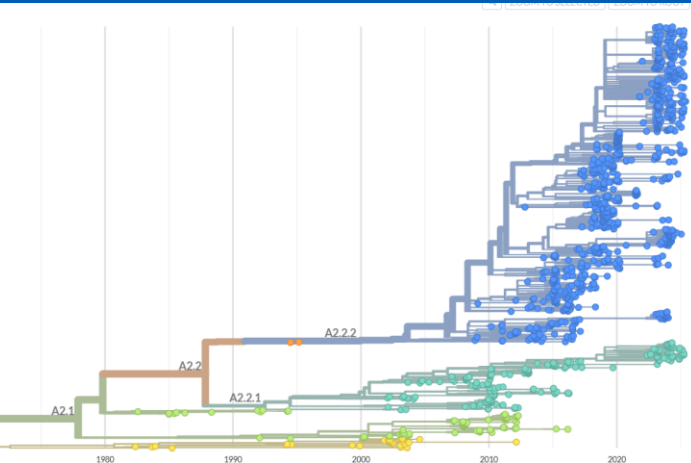




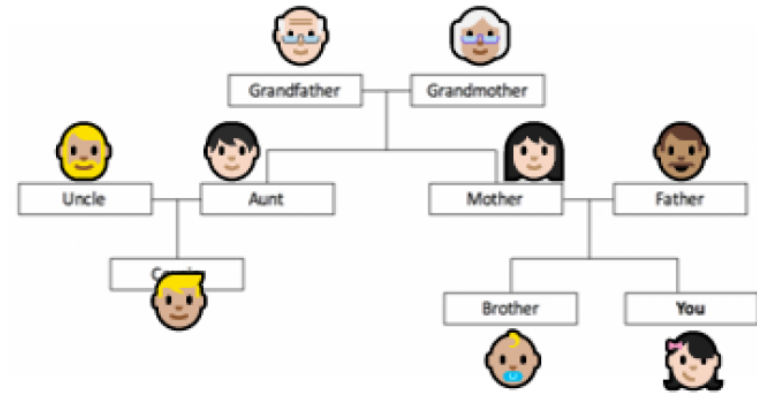
From sequences to trees – a short introduction to phylogenetics

Marie Lataretu



What is phylogenetics?

- The study of **evolutionary relations** among biological entities (populations, organisms, genes)
- Such relationships are almost always inferred from **molecular sequence data**

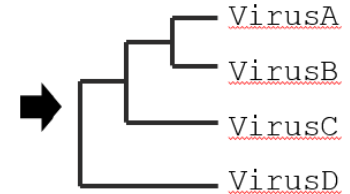


VirusA: CGTTG**C**TGAAAT

VirusB: CGTTGATGAGAA

VirusC: GGT**A**GATGAACG

VirusD: GG**C**TGA**A**GAT**C**T



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Image adapted from Nathan Grubaugh

From Sequences to Trees

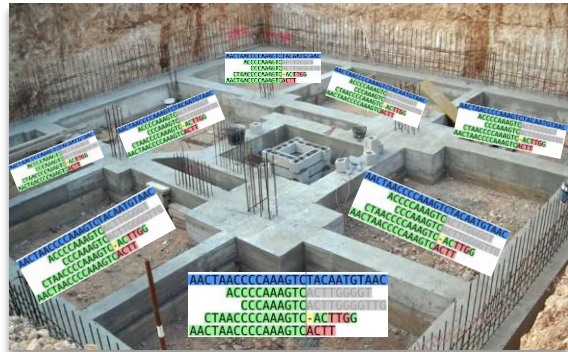
①



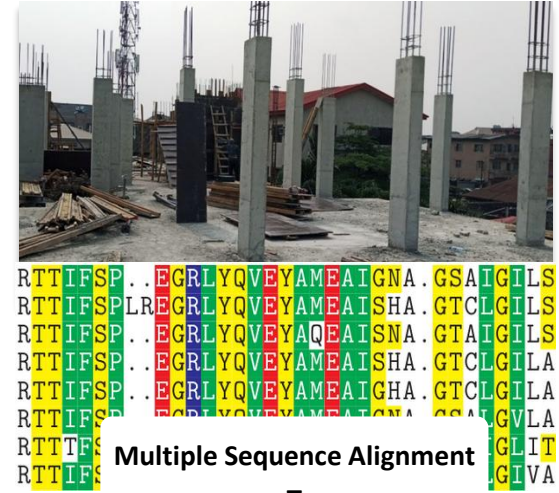
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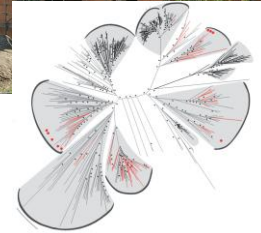
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Consensus Sequences
—
The Foundation



Multiple Sequence Alignment
—
The Columns

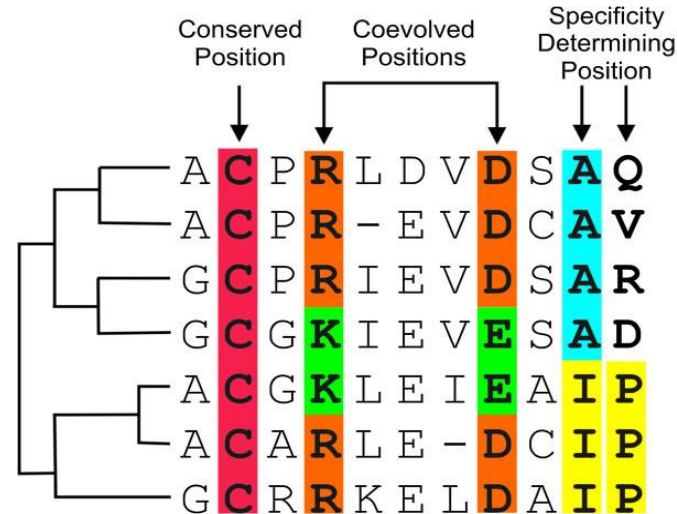


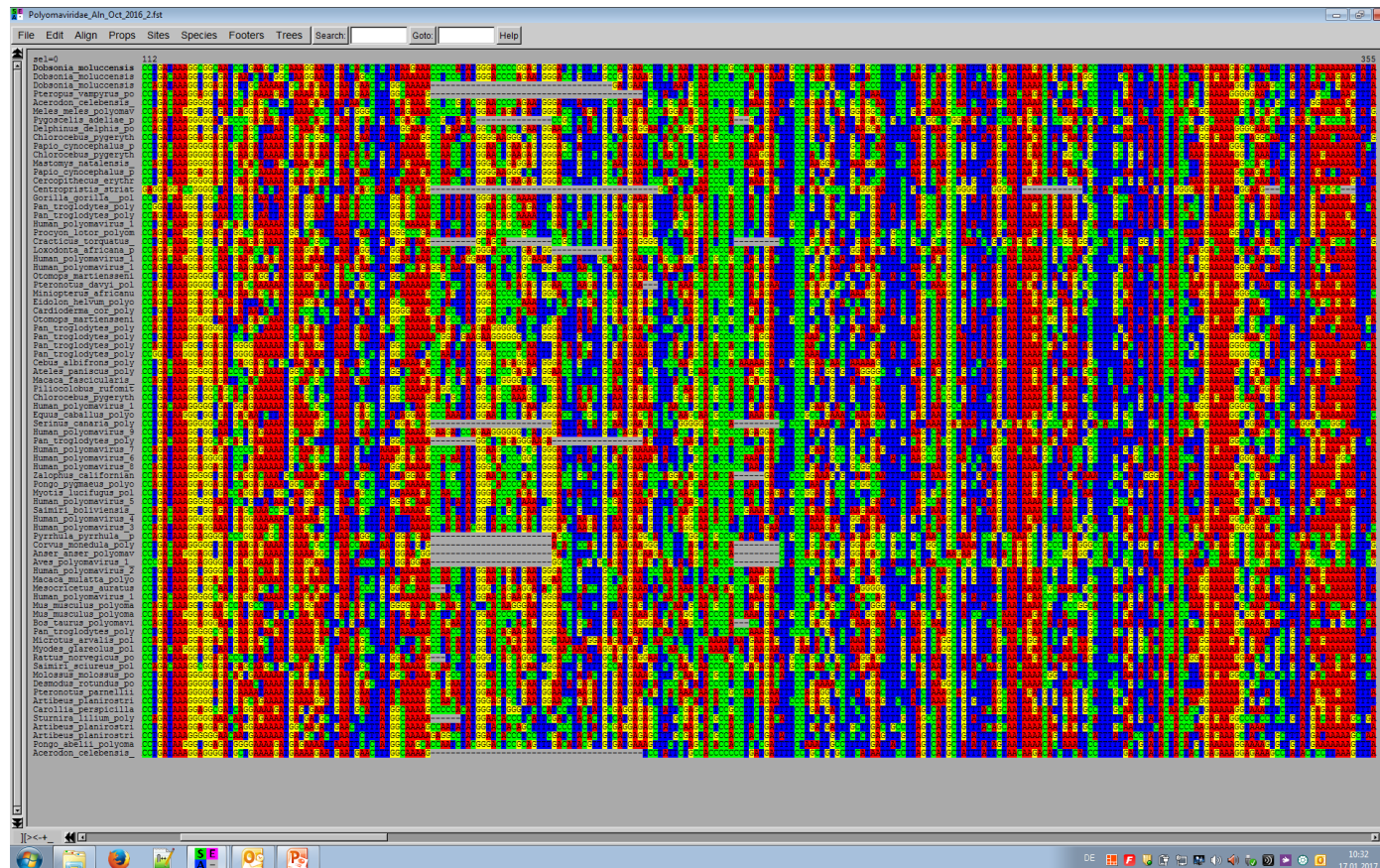
Phylogenetic Tree
—
The Building



Multiple sequence alignment (MSA)

- Columns represent a hypothesis for a common evolutionary history
- Differences infer evolutionary relationships → are then depicted on a phylogenetic tree





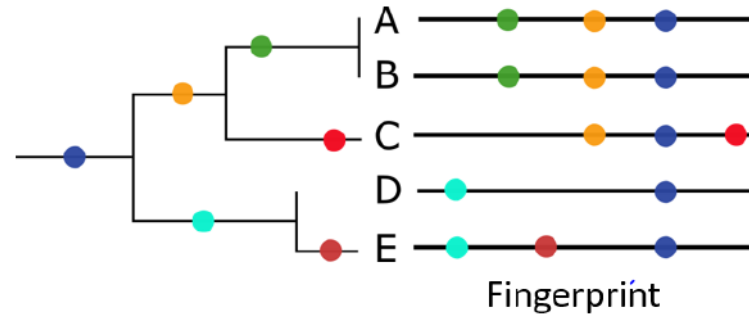


From MSAs to fingerprints

Genetic relatedness (SNP differences)

	2016K-0438	PNUSAS010651	PNUSAS011968	PNUSAS013510	PNUSAS013903	PNUSAS013904	PNUSAS013905
2016K-0438	0	35	37	36	25	28	26
PNUSAS010651	35	0	2	41	31	35	34
PNUSAS011968	37	2	0	41	30	34	33
PNUSAS013510	36	41	41	0	28	33	30
PNUSAS013903	25	31	30	28	0	1	1
PNUSAS013904	28	35	34	33	1	0	2
PNUSAS013905	26	34	33	30	1	2	0

Phylogenetic trees



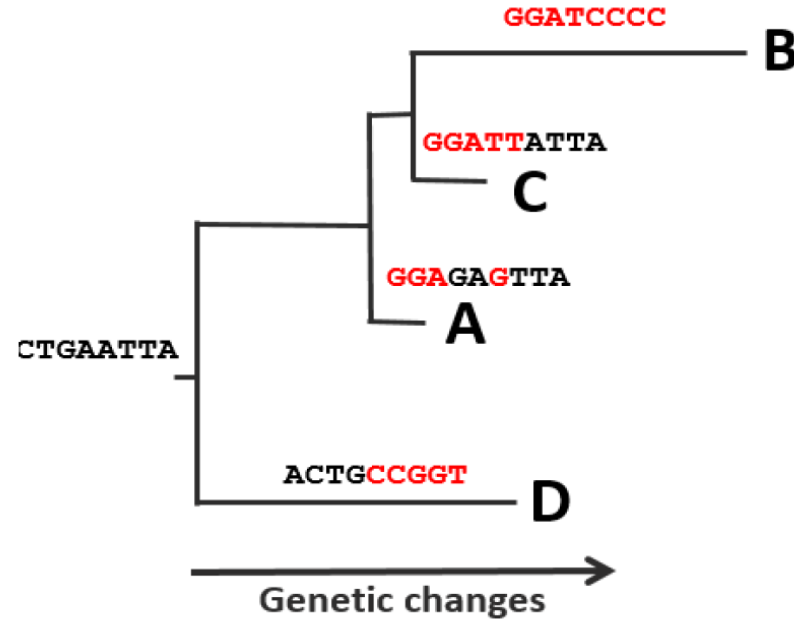
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Tree image from Trevor Bedford Group: <https://docs.nextstrain.org>



Growing trees from MSA

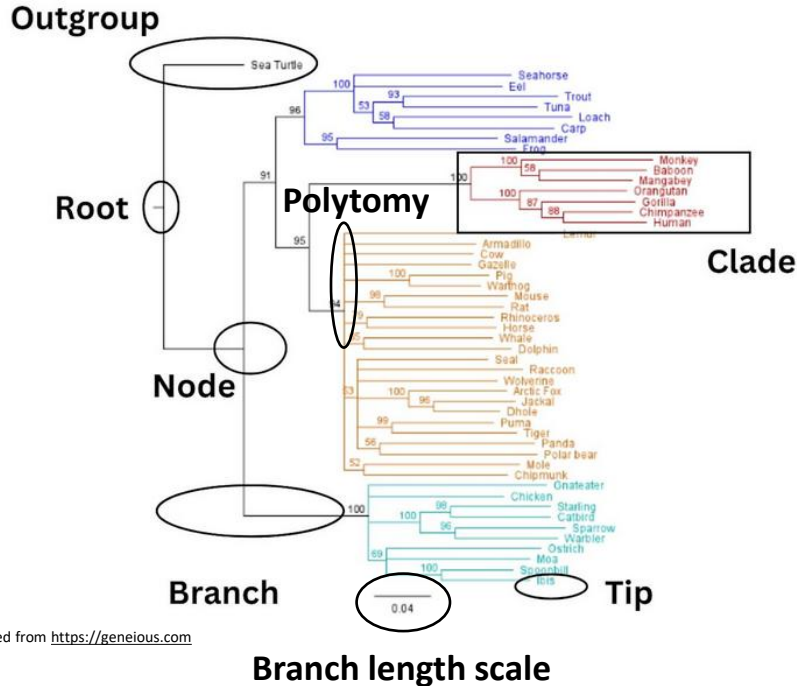
Isolate	Fingerprint
Ancestor	ACTGAATTA
A	GGAGAGTTA
B	GGATCCCCC
C	GGATTATTA
D	ACTGCCGGT



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Anatomy of a Phylogenetic Tree

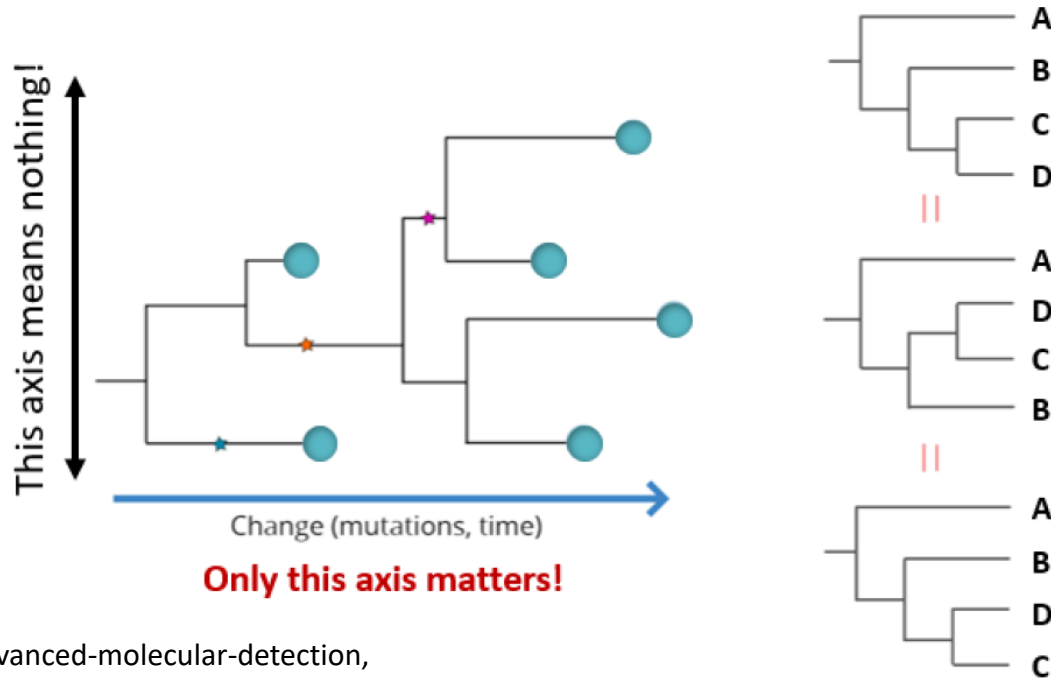


- A tree is a connected graph with **no cycles**
- A **binary tree** has bifurcations in all internal nodes
- Any **two nodes** are connected by a **unique path**
- Internal nodes reflect **ancestral** character states
- **Unresolved nodes** show multifurcations or polytomies
- A **root** gives direction to the tree
- A **weighted tree** has **branch lengths** representing **mutation events**

Image modified from <https://geneious.com>



Anatomy of a Phylogenetic Tree



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Adapted from Robert Bear



Phylogenetic placements

- Place your samples of interest onto a previously-inferred, fixed phylogenetic tree
 - “optimal” position according to the mutations that the samples of interest
 - Frequently used method: placement by means of *parsimony* → least number of *additional* mutations

Tools

USHER: Ultrafast Sample placement on Existing tRee

- USHER (genome.ucsc.edu/cgi-bin/hgPhyloPlace)
 - Focus: genetically similar sequences
 - Finds most closely related samples

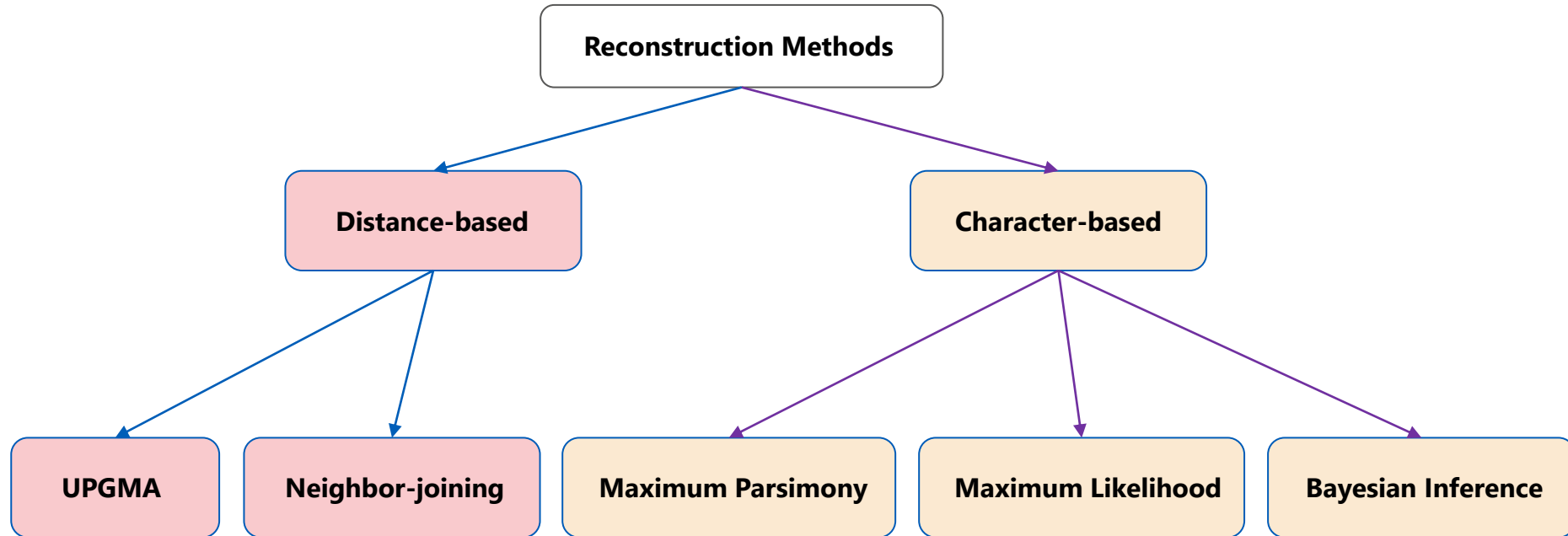
Nextclade^{v3.21.2}

Clade assignment, mutation calling, and sequence quality checks

- Nextclade (<https://clades.nextstrain.org/>)
 - Focus: viral evolutionary history
 - Allows simultaneous placement of multiple samples



Phylogenetic trees/reconstruction



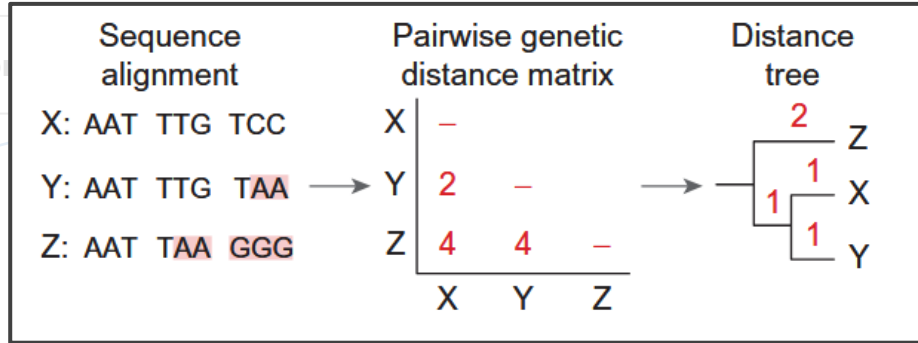


Phylogenetic trees/reconstruction

Distance-based

UPGMA

Neighbor-joining



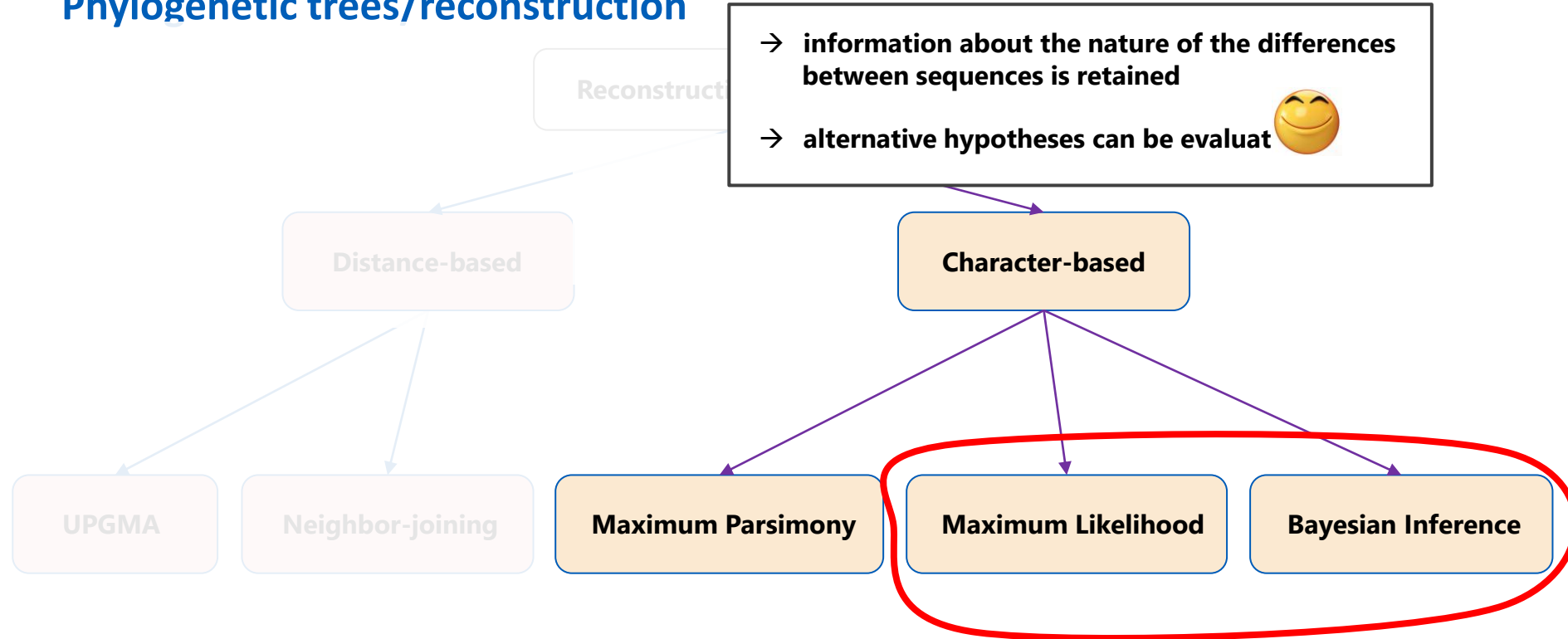
→ information about the nature of the differences between sequences is immediately lost

→ one tree only = limited evaluation of alternative hypothesis





Phylogenetic trees/reconstruction





Phylogenetic placement versus phylogenetic tree versus?

- Phylogenetic placement
 - What sequences are most closely related to mine?
 - Could my outbreak be linked to someone else's?
 - This is the first time I've detected this lineage. Where might it have come from?
 - This person has a travel history. Did their infection result from exposure while traveling?
- → *Targeted question about what other sequences your sequence of interest is most closely related to*
- Phylogenetic tree
 - When did my outbreak start?
 - How long has my outbreak been ongoing?
 - How long has this particular lineage been circulating?
 - Incorporate geographic data
- → *Explore the descriptive epidemiology of an outbreak*



Limitations and implications

- Transmission pathways (and the direction of transmission) cannot be assumed to mirror phylogeny (without other data)
- Causal links (e.g., between cases and exposures) cannot be assumed from sequence data alone
- → Strains that are phylogenetically closer are *more likely* to share an epidemiological association
- → Interpret with caution, all trees are an approximation of the truth!

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Phylogeny vs. Transmission

- Interpret with caution because topology depends on sampling:

